Prob.5. (2&T. ch.8. 1.3)

Consider the "blip" f-n, $F(y)$, like defined in Pr.5. Consider the sol-n to the 2-dim-l plane wave sol-n to wave eq-n:

(1) $$u(x_1, x_2, t) = F\left(\frac{1}{\sqrt{2}}x_1 + \frac{1}{\sqrt{2}}x_2 - t\right).$$

In the $x_1 x_2$ plane draw the lines where the sol-n has its max. value & the strips where $u(x_1, x_2, t) \neq 0$ for $t = 0, 1, 2, 5$.

(?4)

Sol-n:

[Preamble]

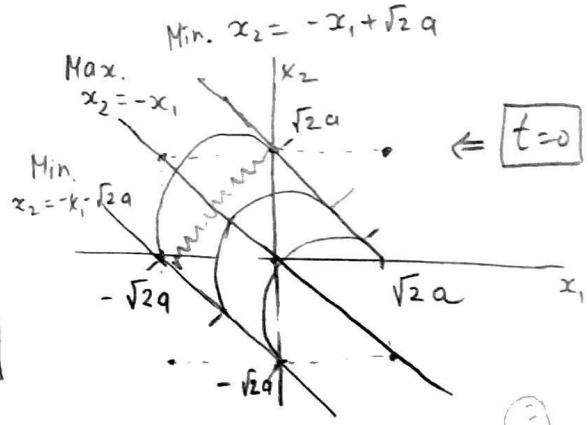
- let's go one step at a time again:

(i) $t = 0$

$$u(x_1, x_2, 0) = F\left(\frac{1}{\sqrt{2}}x_1 + \frac{1}{\sqrt{2}}x_2\right)$$

Max. occurs @ $\frac{1}{\sqrt{2}}x_1 + \frac{1}{\sqrt{2}}x_2 = 0 \Rightarrow \boxed{x_2 = -x_1}$

Min. occurs @ $\frac{1}{\sqrt{2}}x_1 + \frac{1}{\sqrt{2}}x_2 = \pm a \Rightarrow \boxed{x_2 = -x_1 \pm \sqrt{2}a}$

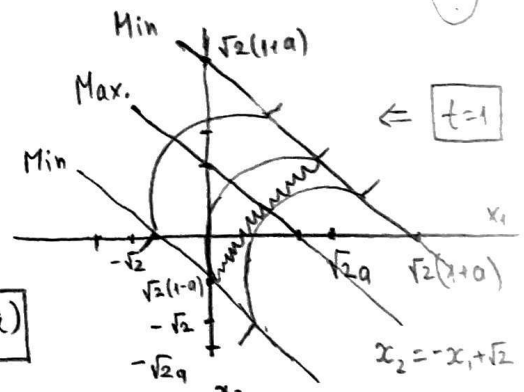


(ii) $t = 1$.

$$u(x_1, x_2) = F\left(\frac{1}{\sqrt{2}}x_1 + \frac{1}{\sqrt{2}}x_2 - 1\right)$$

Max. occurs @ $\frac{1}{\sqrt{2}}x_1 + \frac{1}{\sqrt{2}}x_2 - 1 = 0 \Rightarrow \boxed{x_2 = -x_1 + \sqrt{2}}$

Min. occurs @ $\frac{1}{\sqrt{2}}x_1 + \frac{1}{\sqrt{2}}x_2 - 1 = \pm a \Rightarrow \boxed{x_2 = -x_1 + \sqrt{2}(1 \pm a)}$



(iii) $t = 2$ (TBC)

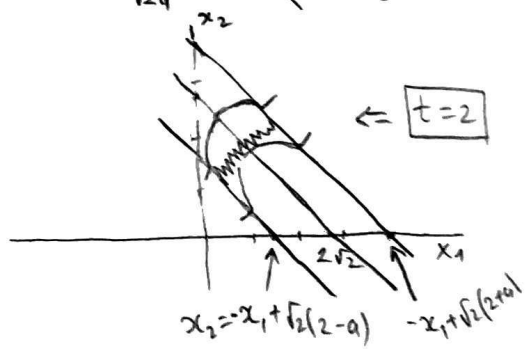
$$u(x_1, x_2, 2) = F\left(\frac{1}{\sqrt{2}}x_1 + \frac{1}{\sqrt{2}}x_2 - 2\right)$$

Max. occurs @ $\frac{1}{\sqrt{2}}x_1 + \frac{1}{\sqrt{2}}x_2 = 2 \Rightarrow \boxed{x_2 = -x_1 + 2\sqrt{2}}$

Min. occurs @ $\frac{1}{\sqrt{2}}x_1 + \frac{1}{\sqrt{2}}x_2 - 2 = \pm a$

$$\Downarrow$$

$$\boxed{x_2 = -x_1 + \sqrt{2}(2 \pm a)}$$



(iv) $t=5$

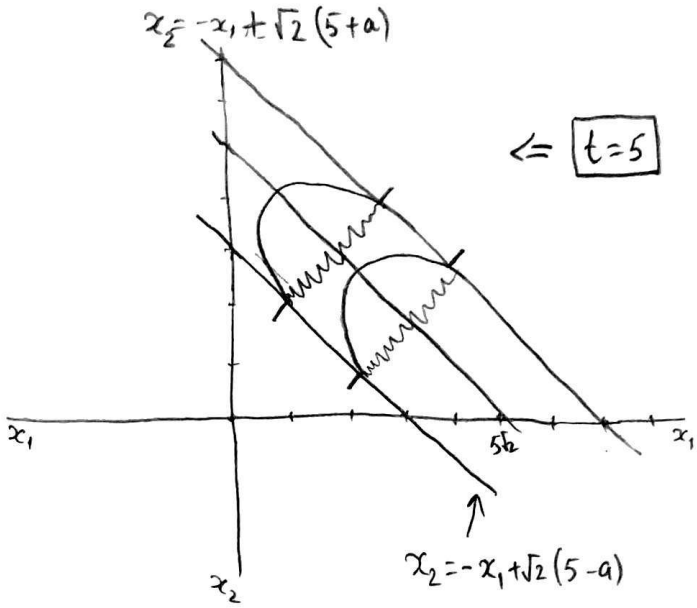
$$u(x_1, x_2, 5) = F\left(\frac{1}{\sqrt{2}}x_1 + \frac{1}{\sqrt{2}}x_2 - 5\right)$$

Max. occurs @ $\frac{1}{\sqrt{2}}x_1 + \frac{1}{\sqrt{2}}x_2 - 5 = 0$

$$\Downarrow$$

$$\boxed{x_2 = -x_1 + 5\sqrt{2}}$$

Min. occurs @ $\frac{1}{\sqrt{2}}x_1 + \frac{1}{\sqrt{2}}x_2 - 5 = \pm a$

$$\Downarrow$$

$$\boxed{x_2 = -x_1 + \sqrt{2}(5 \pm a)}$$